Overall, this project was not without its difficulties. I believe that all five of the applications that we completed throughout the course of this semester have immensely helped in building my understanding of real-time systems. Throughout this project in particular, using application 5, which was a culmination of all the previous applications, really helped to finish solidifying my understanding of a dual-core FreeRTOS pipeline on the ESP32-S3.

In the future, if I were to do anything differently, I would add more realistic details such as a more realistic patient simulator with multiple telemetry profiles to allow the system to switch between the different types of cardiac conditions. This would really help visually cause currently we see all the values in the terminal at once and separating them would help in creating easier to read information. I would also try and include maybe a visual live graph simulation on my main page, if possible, of potential ECG, heart rate, and SpO2 data so that users can visualize exactly how this would work in a real-world context.

For this final application, the hardest part was integrating everything correctly. For this final portfolio I used my application 5 as the basis since it incorporated elements from nearly all the other applications; whiles I understand each individual application, it took a few tries for all the components to work correctly in the scaffold. I believe that application 5/this final project took me the longest of all of the applications within this class. I think part of the reason that I struggled with the portion is because for this project, I had to ensure that the FreeRTOs tasks, queues, evednt groups, direct task notifications, interrupt handling, and the monitoring system were all simultaneously working together without affecting the deterministic execution. To get them all to work and to verify that task timing, measuring WCET, and documenting how each component worked in the system took significantly longer to get correct than anticipated.

The most valuable thing that I learned was how important a deterministic system design is in real-time software. Throughout this course, we've learned a lot conceptually but being able to relate it to real-world events and see how deadlines and small system failures can have fatal consequences has been interesting to learn. I liked the integration of real-world examples cause oftentimes you just learn a concept and can't really visualize it in your everyday life, and I think it was very beneficial the way everything we learned conceptually was applied in a real-world way. I also think this helped in reinforcing my functional code skills as well as giving me more experience with navigating and using GitHub.

To conclude, I really enjoyed all the real-world applications and skills that were learned throughout the course of not only this project, but all the other applications and homework as well. I believe the one skill that I will carry into future projects, and technical interviews is my newfound ability to explain engineering decisions behind my implementation rather than only trying to discuss the code itself. As I stated earlier, this project really strengthened my understanding of FreeRTOS scheduling, dual Core task partitioning, inter-process communication, WCET analysis, interrupt design, hazard mitigation, and many more topics.